Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Victoris
b. Team leader name: Yash Shah (team: Pratyaksh, Krishna)
c. Problem Statement: Sustainability & Social Impact: Second Opinion, a live scam guardian that warns seniors (and every age group) before money moves.

Brief about the idea

Second Opinion is a live scam guardian for the people we love. It listens alongside someone on a suspicious call (speakerphone or a shared Google Meet tab), recognises the coercion pattern behind scams such as the fake-police 'digital arrest', and warns them calmly in English or Hindi BEFORE any money moves.

It then brings in a family guardian, requests a cool-off hold and drafts the cybercrime complaint (1930 / cybercrime.gov.in). The same engine also checks screenshots, voice notes, pasted messages or a plain question ('is this a scam?'), and extends from seniors to teens and young adults.

Why it matters: most losses are payments the victim authorises under pressure on a call, which transaction monitoring cannot see.

- Japan 2025: JPY 142.3B lost to special fraud (+98%); people 65+ bore 59.2%.
- India 2024: INR 1,935 crore lost to 'digital arrest' scams (1.23 lakh complaints).
- Australia 2025: people 65+ are 17.1% of the population but 26.5% of reported losses.

Theme: Sustainability & Social Impact | Category: Social Good

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How different is it from existing ideas?

- Existing tools check transactions after the fact, filter spam, or analyse a single uploaded message. Second Opinion works DURING the call, reasoning about the coercion state (authority + urgency + secrecy + payment request), not keywords, in English / Hindi / Hinglish.
- It acts: guardian alert, guardian-approved cool-off hold, ready-to-file complaint draft. Every alert shows the exact quote that triggered it.

How will it solve the problem?

- It interrupts the scam at the one moment it can be stopped: while the victim is on the call and before they authorise the transfer.
- A family member is pulled in within seconds, breaking the isolation scammers rely on. Reporting is instant (feeds India's 1930 'golden hour').

USP

- Live, multilingual, explainable (quotes + official advisories), human-in-the-loop (guardian approves), privacy-first (no audio stored), and ONE engine for every age group: seniors today, teens (sextortion) and young adults (job/task scams) next.

List of features offered by the solution

1. Call Guard: listens to speakerphone, a Google Meet tab or a demo call; live transcript; risk meter (Calm / Caution / Warning / STOP).
2. Signal timeline: each scam signal with the exact quote and timestamp (authority claim, urgency, secrecy, payment/OTP request, threat ...).
3. Whisper warning in English or Hindi and a one-tap 'Hang up & call 1930' button; large, calm, senior-friendly UI.
4. Case agent (Gemini function calling): checks the official advisory, alerts the guardian, requests a cool-off hold, drafts the complaint.
5. Family guardian console on a second phone (QR / pairing code): realtime alerts, approve or decline the hold.
6. Complaint draft for 1930 / cybercrime.gov.in: category, summary, timeline, quotes, amount requested; copy / print / PDF.
7. Scam Check: upload chat screenshots or audio recordings, paste a message/link text, or ask a question; verdict, risk score, red flags, safe reply, follow-up chat.
8. Frame Check: periodic video-call frame checked for staged authority cues (uniform, fake warrant) - cues only, never a deepfake verdict.
9. All-ages profiles: Senior (core), Adult, Young adult, Teen (beta) change tone and scam taxonomy; Hindi / English interface.
10. Impact page: sourced statistics and an avoidable-loss calculator (12% conservative to 94% lab ceiling).
11. Recorded Replay mode and evaluation harness (16 scripted calls: 10 scam, 6 genuine).

Process flow diagram or Use-case diagram

LIVE CALL PATH

1. Call audio
Speakerphone mic, shared Google Meet tab, or demo call. 16 kHz PCM from the browser.

→

2. Gemini analysis
8-second audio slices to Gemini Flash-Lite: transcript + scam signals + exact quotes. (Optional Gemini Live streaming.)

→

3. Risk meter
Calm / Caution / Warning / STOP, escalating with hysteresis. Meet video frame checked for staged-authority cues.

→

4. Case agent
Gemini Flash function calling: official advisory, guardian alert, cool-off hold, complaint draft.

Genuine call
No signals: stays Calm. Nothing is raised, nothing is stored.

Warning
Calm whisper in English / Hindi + "Hang up and call family" button.

STOP
Guardian phone alerted (Firestore realtime). Guardian approves the cool-off hold.

Report
Complaint draft for 1930 / cybercrime.gov.in with timeline and quotes. Nothing is filed automatically.

SCAM CHECK PATH

Input
Screenshot, voice note / call recording, pasted text, or a question. Profile: senior / adult / young adult / teen.

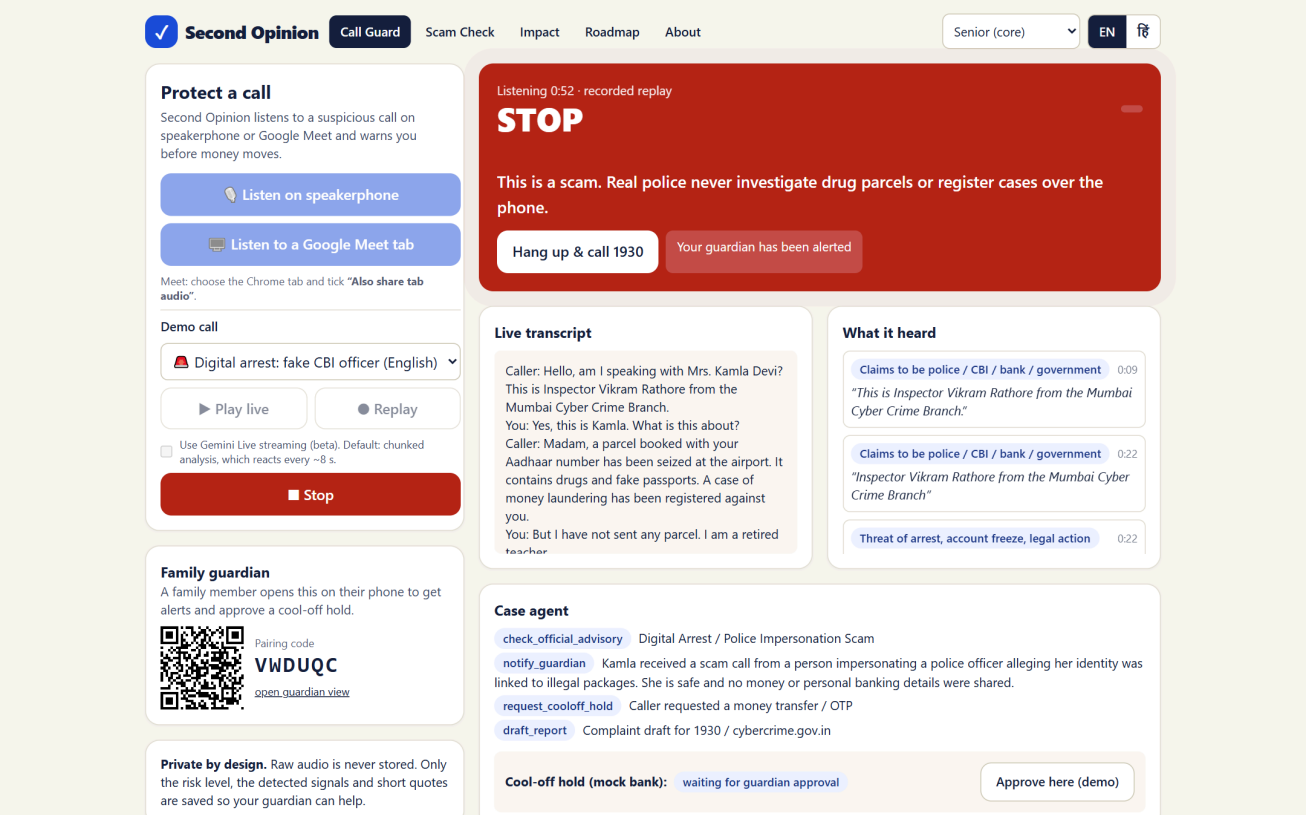
→

Gemini multimodal
Structured JSON verdict: risk 0-100, scam type, red flags with quotes.

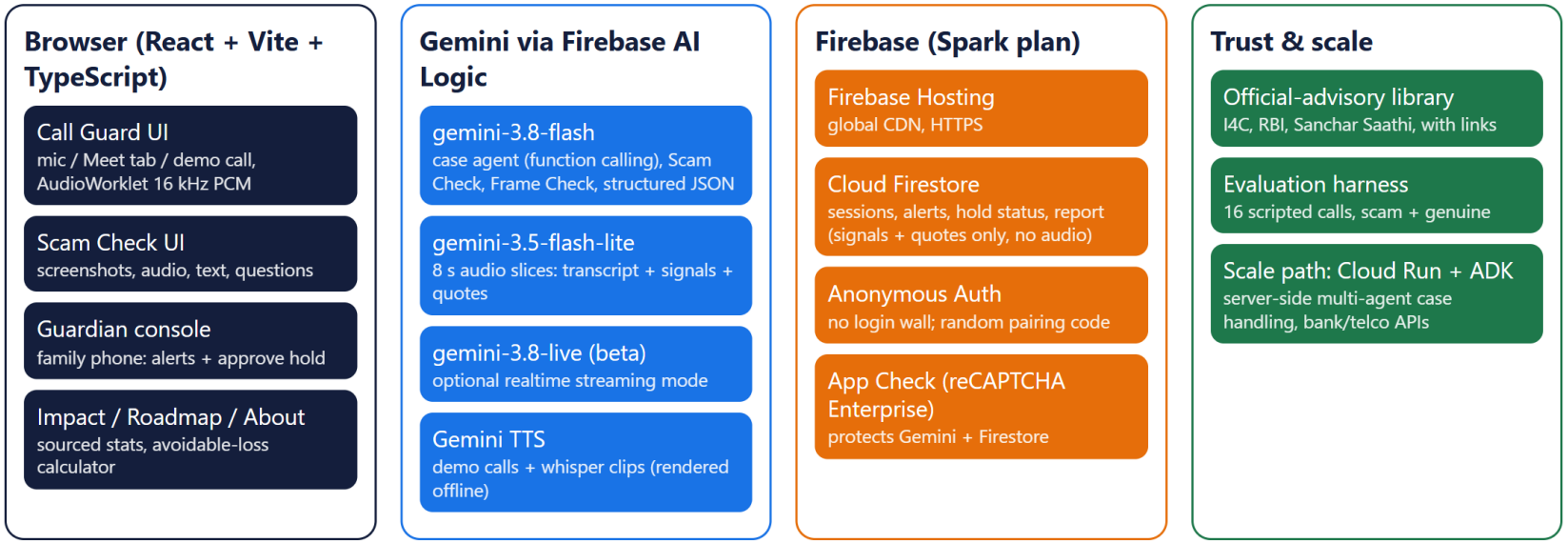
→

Guidance
What to do now, safe reply, official advisory links, helplines, follow-up chat.

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Data flow: browser audio/images → Firebase AI Logic (App Check verified) → Gemini → structured alerts → Firestore → guardian phone. Rules: only the session owner writes; a guardian may change only the hold status.

Technologies to be used in the solution

Google AI / Cloud

- Gemini via Firebase AI Logic: gemini-3.8-flash (case agent with function calling, Scam Check multimodal + structured JSON, Frame Check vision), gemini-3.5-flash-lite (8-second audio slices: transcription + scam signals), gemini-3.8-live (optional realtime streaming, beta), Gemini TTS (demo audio).
- Firebase Hosting (deployed prototype), Cloud Firestore (realtime guardian sessions), Firebase Anonymous Auth, Firebase App Check (reCAPTCHA Enterprise).
- Scale path on Google Cloud: Cloud Run + Agent Development Kit for server-side multi-agent case handling; Agent Platform memory; Gemma 4 on-device.

Application

- React 19, Vite, TypeScript, Tailwind CSS; Web Audio API (AudioWorklet, 16 kHz PCM), getDisplayMedia for Meet tab audio and video frames.
- Python tools (google-genai): TTS rendering of demo calls, recorded-replay builder, evaluation harness, deck/submission builder.

Safety & privacy

- Curated official-advisory library (I4C, RBI, Sanchar Saathi); quotes for every signal; guardian approval; no raw audio or video stored; Firestore rules: only the session owner writes, a guardian can change only the hold status.

Estimated implementation cost (optional)

Prototype: USD 0. Firebase Spark plan (Hosting, Firestore, Auth, App Check) + Gemini API free tier.

Production estimate (assumptions, not a quote): one protected 10-minute call is about 75 audio slices of 8 s (~860 input tokens and ~150 output tokens each) = ~65k input + ~11k output tokens.

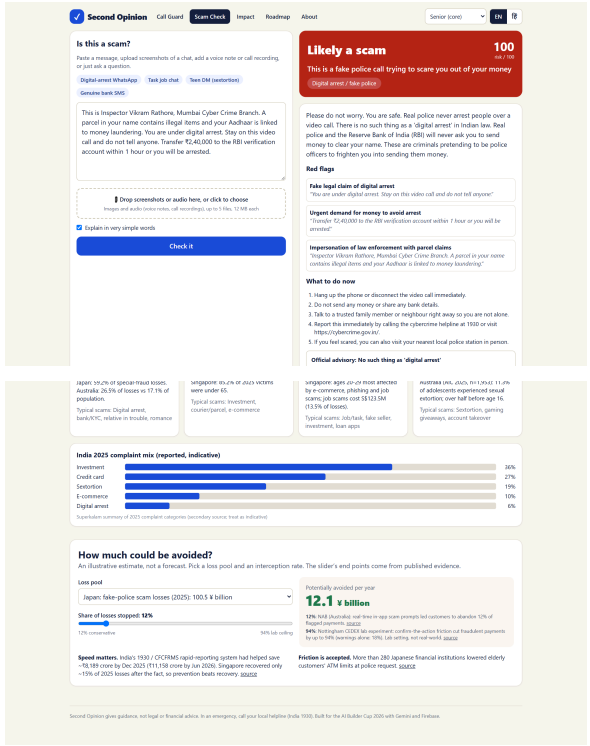
- At Gemini 3.8 Flash intro prices (USD 0.75 / 3.75 per 1M tokens) that is about USD 0.09 per 10-minute call; Flash-Lite is cheaper.
- Case agent + report: a few extra calls, about USD 0.01 per escalated call.
- Firestore / Hosting: well under USD 0.01 per call at this scale.

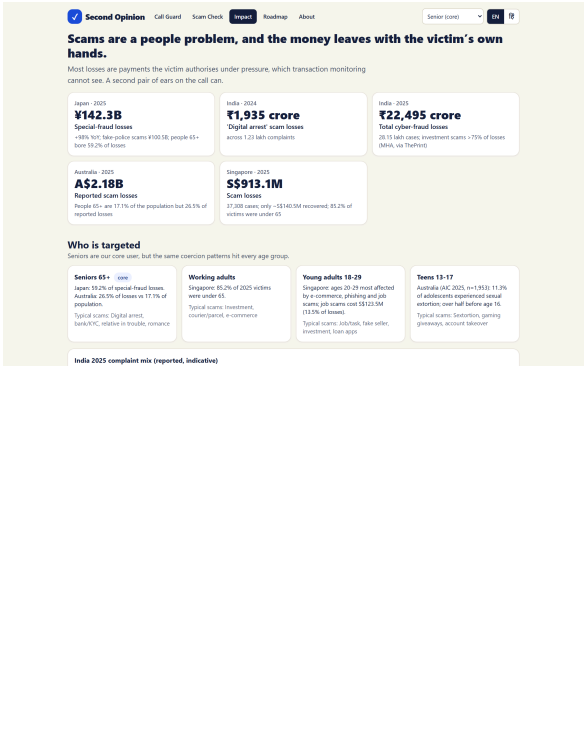
So protecting 10,000 calls a month costs roughly USD 1,000 or less, against average per-victim losses that are orders of magnitude higher (e.g. INR 1,935 crore across 1.23 lakh digital-arrest complaints is about INR 1.6 lakh per complaint).

Scale costs fall further with on-device Gemma 4 for first-line detection and server calls only for escalations.

Snapshots of the prototype







Live prototype: Call Guard (STOP + case agent + guardian), Scam Check verdict, Impact page. https://gen-lang-client-0095149296.web.app

Prototype Performance report/Benchmarking

Evaluation (gemini-3.5-flash-lite), 16 scripted calls run through the SAME classifier prompt the app ships:

- Scam calls detected (warning or stop): 100% of 10 | False alarms on genuine calls: 0% of 6
- Median estimated time to first warning: 10.0 s (estimated from word counts at ~2.4 words/s)

Measured on the deployed prototype:

- English digital-arrest demo call (audio): STOP with correct quotes at ~24 s; guardian alert + complaint draft within seconds after that.
- Scam Check (text, low thinking level): result in ~9 s; deployed on Firebase Hosting with App Check.

Caveat: scripted, small-scale and indicative - not a field trial.

Call	Type	Peak	First warning	Call	Type	Peak	First warning
digital-arrest-en	scam	stop	24s	jp-fake-police	scam	stop	0s
digital-arrest-hi	scam	stop	20s	jp-oreore	scam	stop	0s
digital-arrest-hinglish	scam	stop	10s	genuine-bank-callback	genuine	calm	-
bank-kyc-otp	scam	stop	10s	genuine-doctor	genuine	calm	-
relative-in-trouble	scam	stop	8s	genuine-grandson	genuine	calm	-
tax-refund	scam	stop	22s	genuine-delivery	genuine	calm	-
investment-guaranteed	scam	stop	24s	genuine-insurance-renewal	genuine	calm	-
courier-customs	scam	stop	8s	genuine-electricity	genuine	calm	-

Additional Details/Future Development (if any)

From one grandparent's phone call to every age group

- Teen Shield: sextortion, gaming-currency and giveaway scams; non-judgmental tone; risk-only alerts to a trusted adult (never the teen's private content).
- Young adults & merchants: job/task scams, fake sellers, loan apps, fake payment screenshots and UPI collect-request traps.
- Bank & telco integration: a real cool-off hold API with banks, call-risk metadata from telcos, auto-reporting to Sanchar Saathi / 1930.
- On-device Gemma 4 (E2B/E4B): private offline detection, Android call-screening app, WhatsApp bot, Meet / WhatsApp Web extension.
- Production backend on Google Cloud: Cloud Run + ADK multi-agent case handling, Agent Platform memory for per-family patterns, observability.
- More JAPAC languages via Gemini Live translation (Tamil, Bengali, Bahasa, Japanese, Korean ...).
- Gemini Live streaming for sub-second reactions once turn-detection can be tuned for continuous call audio.

Honest limits today: demo calls and speakerphone/Meet audio only (phones block call-audio capture); free-tier Gemini data may be used by Google to improve products, so a production version needs paid-tier or Vertex AI data controls and consent flows; evaluation is scripted, not field-tested.

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub Public Repository:
https://github.com/dager23/second-opinion

Demo Video Link:
(add demo video link)

Final Product Link:
https://gen-lang-client-0095149296.web.app

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Thank you

